

ICT1011: Computer Organisation and Architecture

Tutorial 5: Executable Assembly Instructions

Q1. Arithmetic Operations Instructions

- (a) Write an assembly program to help you subtract two 32-bit numbers:
0x01234567 - 0x89ABCDEF
and store the result in two registers, R11 storing bits 0-15 of the results and R12 storing bits 16-31 of the results?

Q2. Logical and Bit Manipulation Instructions

- (a) You are required to complement bit 15 and bit 0 in register R10. For example, if the current value in R10=0xFF00, your operation will change it to R10=0x7F01. Give a single instruction to achieve this requirement in the most optimized way possible.
- (b) Give four different instructions to clear the word value in register R10 to 0x0000.

Q3. Rotation Instructions

- (a) You are required to write a segment of code to multiply the signed byte-sized data in R10 by 5 and leave the result as a word in register R10. Provide two different solutions for this, one utilising the rotation instructions and the other without. Give the total number of bytes and states required to realize this arithmetic computation in each case.
Hint: Note that $(y = a \times 5)$ is equivalent to $(y = a + a + a + a + a)$ or $(y = (a \times 4) + a)$.
- (b) Write a short segment of code that will turn the word-sized value in register R9 to 1 (i.e. 0x0001) if its initial word-size content is negative or to 0 (i.e. 0x0000) if its initial word-sized content is positive. You are not allowed to use any of the program control instruction in your solution.